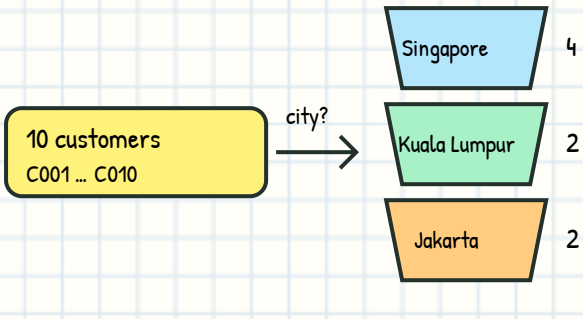
Grouping · my notes L11

sort the items into buckets, then do something with each bucket

groupingBy = buckets 🪣



downstream = what each bucket becomes:

counting() → {Bangkok=1, …, Singapore=4}

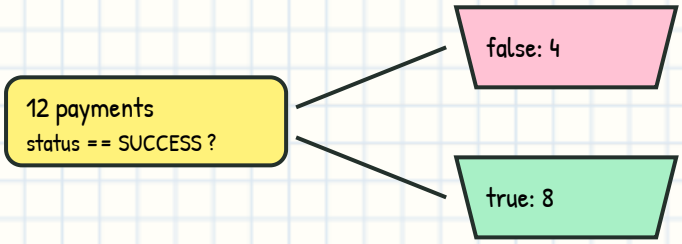
mapping(id, toList()) → [C001, C003, C005, C010]

reducing(ZERO, amount, add) → BigDecimal

default map = HashMap → order?? use TreeMap

+ Bangkok 1, Colombo 1

partitioningBy = exactly 2 buckets ⚖



empty bucket?
still there: true=0

filter vs filtering [9] 🔍

filter first → {PAYNOW=1, PAYPAL=1}

CARD vanished!

filtering inside →

{CARD=0, PAYNOW=1, PAYPAL=1}

Optional per bucket 🎁

maxBy → Optional[T018]

collectingAndThen(…, orElseThrow)

→ ACC-1001=T018 (a bucket is never empty)

Pick the map 🗺

TreeMap::new → Bangkok … Singapore

EnumMap → SILVER, GOLD, PLATINUM

default HashMap → no order

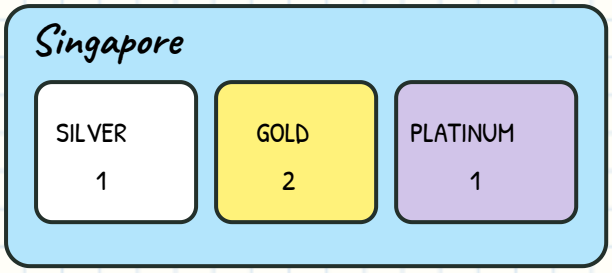
Money per bucket 💰

reducing(ZERO, amount, add)

TRANSFER → 10362.65

summingDouble ✕

Buckets inside buckets



nested: groupingBy(city, groupingBy(tier, counting()))

Singapore={SILVER=1, GOLD=2, PLATINUM=1}

flat: groupingBy(c -> new CityTier(city, tier))

Singapore/GOLD=2 · easy to sort and export

`mapping(name, joining(", "))` → names per tier

`flatMapping` [9] → accounts per tier

`teeing` [12] → CARD=5/2070.29

`groupingBy(YearMonth)` → 2026-11=[POL-502, POL-505, POL-509]

group, then sort the entries → [TRANSFER=15, DINING=10, SALARY=8]

sum of all groups == grand total ✓ (tested)

Worked example: debit report per category

BEFORE: HashMaps + if-null loops · AFTER [8]: 3 groupingBy passes · MODERN: one `groupingBy(category, EnumMap, collectingAndThen(toList(), CategoryStats::of))` · TRANSFER: 8 debits, total 7190.92, largest T042 1432.93 … DINING: 10 debits, 839.28 · all 3 versions agree ✓

Exercise: 12 grouping problems 🧩

avg premium {LIFE=2125.00, HEALTH=1032.88, MOTOR=1275.25, TRAVEL=197.50, HOME=782.63} · richest account per customer · top-3 per account · FAILED per method with zeros

★ Remember

classifier + downstream · pick the map · filter removes keys, filtering keeps them · partitioningBy always has 2 keys · unwrap the Optional inside · next → L12 primitive streams